



Science

Stage 8

Paper 1

2023

Cambridge Lower Secondary Progression Test

Name

Class

Date

45 minutes

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Write your answer to each question in the space provided.
- You should show all your working on the question paper.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

- 1 Draw a straight line to match each **blood component** to its correct **function**.

Draw three lines only.

blood component

function

red blood cell

carries carbon dioxide

white blood cell

fighting pathogens

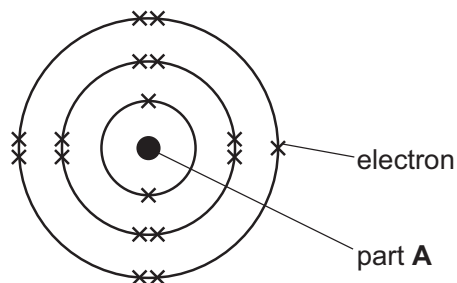
plasma

digests food

transports oxygen

[2]

- 2 Look at the model of a diagram of an atom.



- (a) Write down the name of part **A**.

..... [1]

- (b) Which **two** particles make up part **A**?

..... and [1]

- (c) Which scientist developed the nuclear or planetary model of the atom?

..... [1]

- (d) Write down the charge on an electron.

..... [1]

- (e) The electrons in an atom are held in position.

They do **not** escape from the atom.

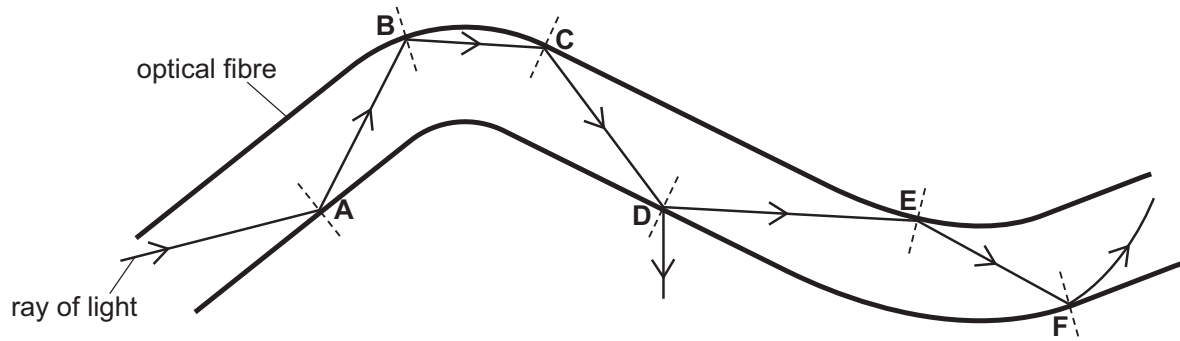
Explain how electrons are held in position.

Use ideas about the charges on the particles.

.....

 [2]

- 3 Mia draws the path of a ray of light through an optical fibre.



A ray of light travels through an optical fibre without any of the ray escaping outside of the fibre.

Mia makes **three** mistakes in her drawing.

Complete the sentences to describe her mistakes at **B**, **D** and **F**.

The mistake at **B**

.....

The mistake at **D**

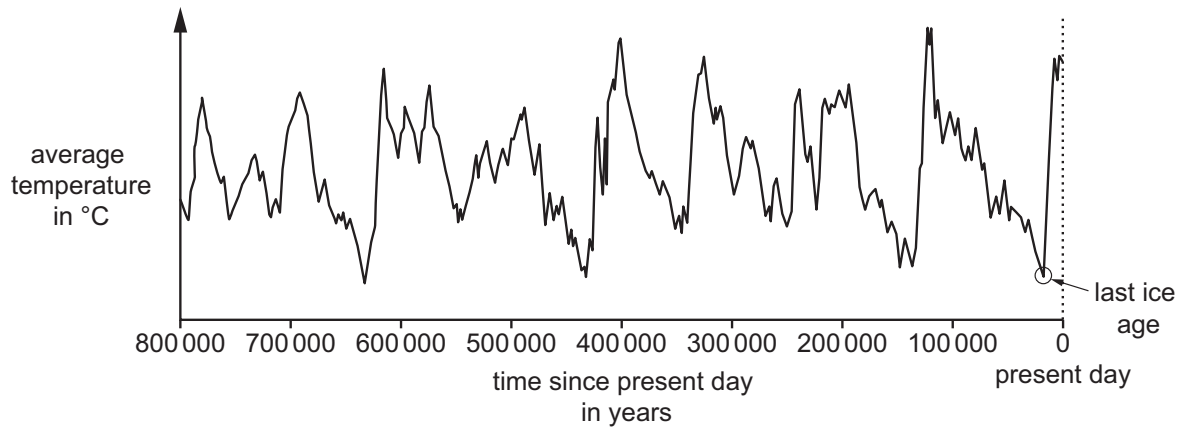
.....

The mistake at **F**

.....

[3]

- 4 Look at the graph showing how the average temperature of the surface of the Earth has changed over the last 800 000 years.



- (a) The last ice age is circled on the graph.

Circle a point on the graph when **one other** ice age happened.

[1]

- (b) Describe the pattern shown in the average temperature of the surface of the Earth over the past 800 000 years.

.....

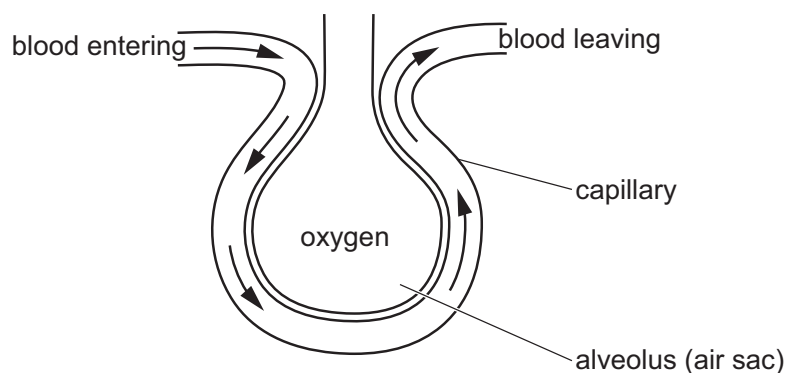
 [1]

- (c) Changes in the percentage of some gases in the atmosphere cause climate change.

Name **one** gas that causes climate change.

..... [1]

- 5 (a) Look at the diagram showing a blood capillary carrying blood to and from an alveolus (air sac) in the lungs.



- (i) Oxygen in the alveolus (air sac) enters the blood.

Write down the name of this process.

..... [1]

- (ii) Blood contains dissolved carbon dioxide.

Describe what happens to the carbon dioxide in the blood capillary.

.....
 [1]

- (b) The table shows the percentage of gases in air as it is breathed in and breathed out.

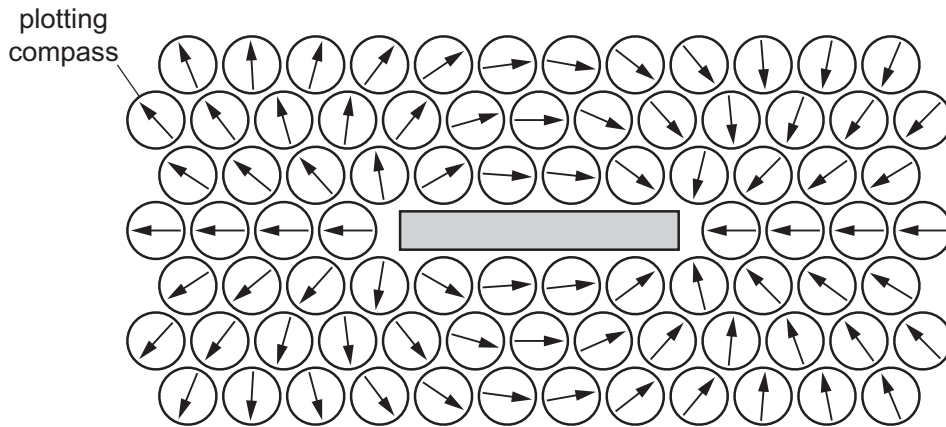
gas	breathed in %	breathed out %
carbon dioxide	0.04	
other gases	78.96	78.96
oxygen	21.00	

Predict the percentages of carbon dioxide and oxygen in **breathed out** air.

Write your answers in the table.

[2]

- 6 Carlos puts plotting compasses around a magnet.



- (a) Look at the plotting compasses.

Describe what the plotting compasses show.

.....

.....

..... [2]

- (b) The magnet is in the middle of the plotting compasses.

Which diagram shows the position of the poles of the magnet?

Circle the correct answer.

N S

N N

S N

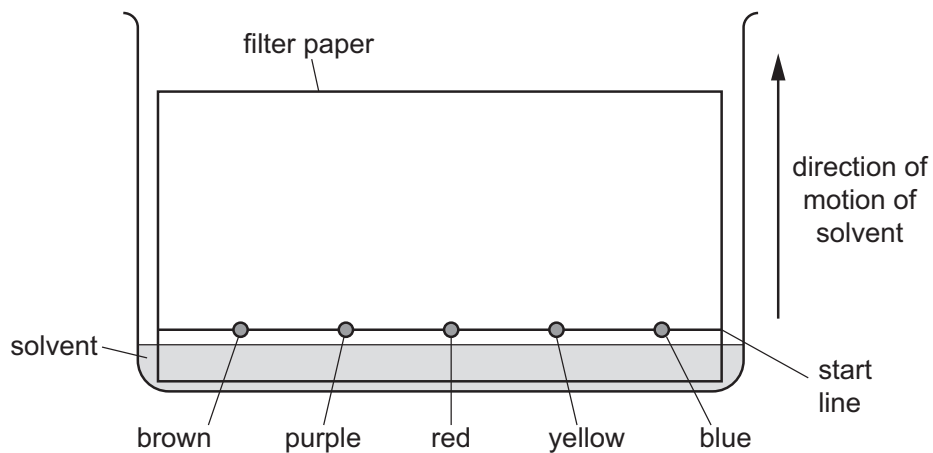
S S

[1]

7 Oliver separates some food colourings.

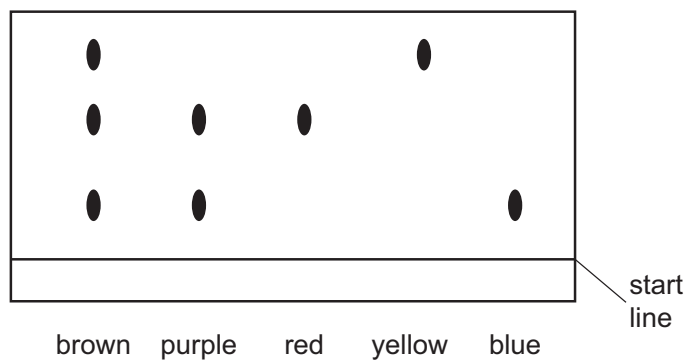
Oliver:

- draws a start line in pencil on some filter paper
- places dots of different food colourings on the start line
- puts the filter paper in the solvent as shown in the diagram



- waits until the solvent has soaked up the filter paper.

Look at his results.



(a) Write down the name of this separation technique.

..... [1]

(b) Which food colourings are **not** pure substances?

.....

Explain your answer.

.....

.....

[2]

(c) Write down the colours in the purple food colouring.

Choose from the list.

brown

red

yellow

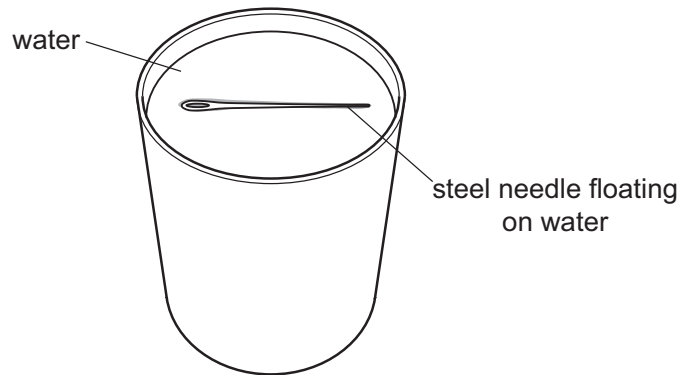
blue

..... [1]

8 Angelique investigates a steel needle floating on water.

In her first experiment Angelique:

- passes a steel needle over a strong magnet
- floats the steel needle on the surface of the water.



She repeats the experiment five times.

Each time, she floats the steel needle in different directions on the surface of the water.

The needle turns to point in the same direction every time.

Explain why.

.....

.....

.....

..... [2]

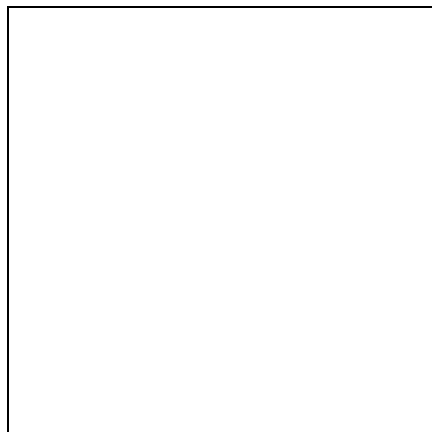
- 9 Pure substances contain either one element **or** one compound.

Impure substances are mixtures.

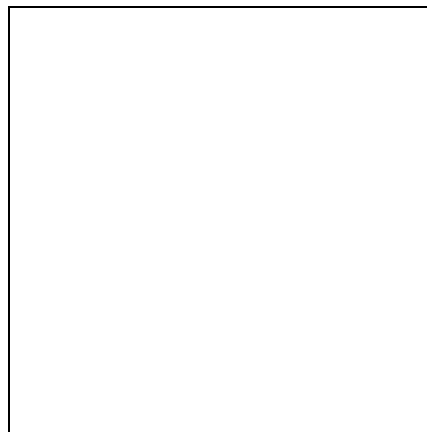
- (a) Draw diagrams to show a model of the molecules in pure water **and** in impure water.

A molecule of water is represented by ○

A molecule of sugar is represented by ●



pure water



impure water

[2]

- (b) Which word describes using a model?

Circle the correct answer.

analogy

conclusion

precision

symbol

[1]

10 Safia plays a game of tennis.

(a) Look at the table showing the percentages of three nutrients in four foods.

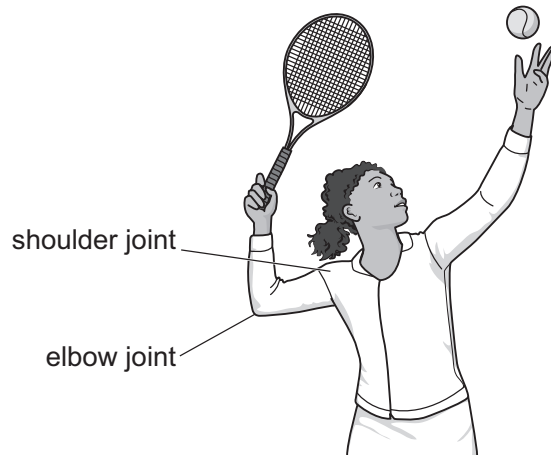
food	protein %	carbohydrate %	fat %
banana	4	88	2
boiled egg	13	1	11
cola drink	0	10	0
chicken sandwich	19	24	6

To give her energy, Safia eats 100 g of one of the foods before she plays her game of tennis.

Which food in the table contains the most energy?

..... [1]

(b) When Safia hits the ball, her shoulder joint and her elbow joint move.



(i) Write down the name of the type of joint at the shoulder.

..... [1]

(ii) Explain how antagonistic muscles allow movement around the elbow joint.

.....
 [1]

(c) Three of Safia's friends make predictions about her breathing rate.

I predict her breathing rate at the start will be **lower** than at the end of the game.



Lily

I predict her breathing rate at the start will be **higher** than at the end of the game.



Gabriella

I predict her breathing rate at the start will be the **same** as at the end of the game.



Mike

Circle which friend has the correct prediction.

Lily

Gabriella

Mike

Explain your answer.

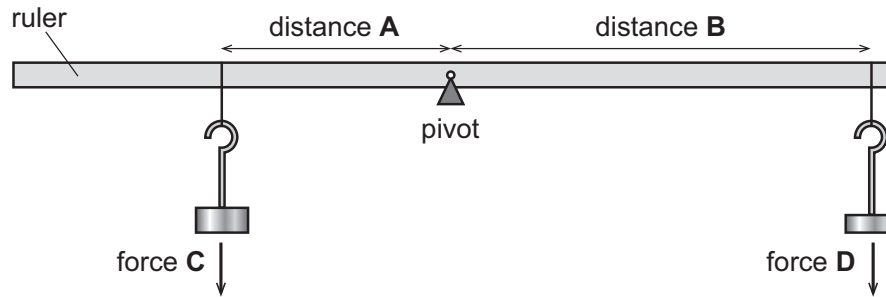
.....

.....

[2]

11 Chen investigates turning forces.

He balances a ruler using different forces and distances.



(a) The turning force is called a moment.

Describe how to calculate a moment.

.....

..... [1]

(b) Chen makes the ruler balance three times.

He writes down the different measurements each time.

distance A = 6 cm, distance B = 12 cm, force C = 4 N, force D = 2 N
distance A = 5 cm, distance B = 2 cm, force C = 4 N, force D = 10 N
distance A = 4 cm, distance B = 8 cm, force C = 2 N, force D = 1 N

(i) Complete the table using these results.

distance A in cm	force C in N	distance B in cm	

[3]

(ii) Chen says,

‘I think my results are reliable.’

Tick (✓) to show if Chen is correct.

Yes ☐ No ☐

Explain your answer.

.....

.....

[1]

12 This question is about energy resources.

(a) Look at the list of resources.

bioplastics

fossil fuels

tidal power

vegetable oil

solar power

Complete the table to group the resources into renewable **and** non-renewable resources.

renewable	non-renewable

[2]

(b) Wind power and coal are resources used to generate electricity.

Describe **two** advantages of using wind power instead of coal to generate electricity.

1

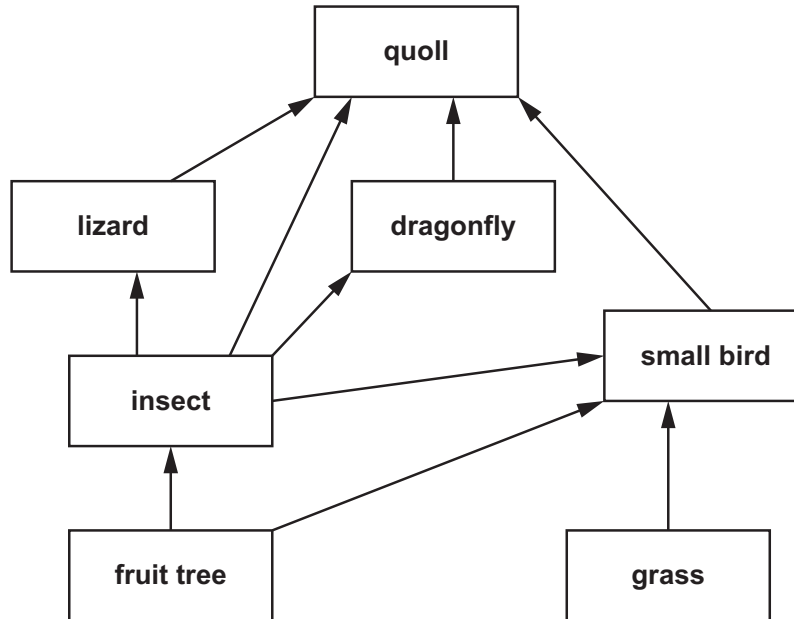
.....

2

.....

[2]

13 Look at the food web for an Australian grassland ecosystem.



(a) Cats are an invasive species in Australia.

Cats eat small birds and lizards.

Explain the effect the introduction of cats has on the dragonfly population in this food web.

.....

.....

.....

..... [2]

(b) Insects eat fruit.

Farmers use a toxic substance to kill insects that eat the fruit on their fruit trees.

Quolls are also killed by this toxic substance.

Explain why.

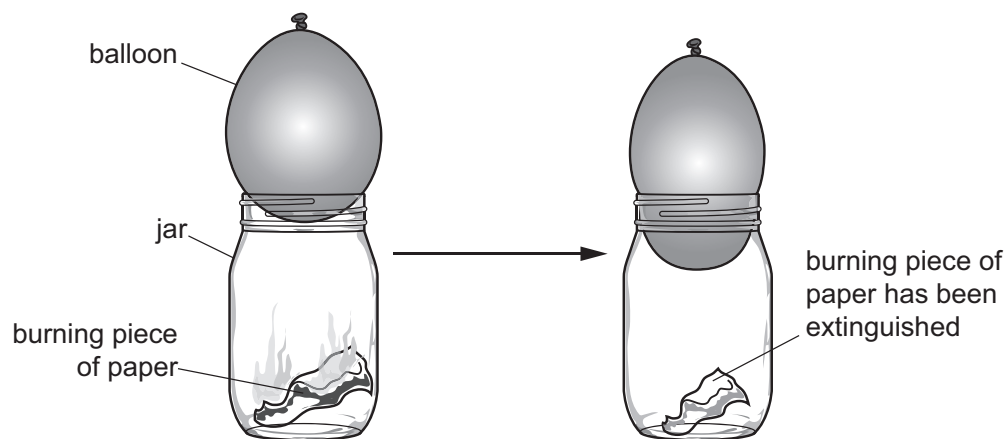
.....

..... [1]

14 A teacher investigates pressure.

The teacher:

- fills a balloon with water
- places a burning piece of paper in a jar
- puts the balloon on top of the jar.



(a) Describe how the teacher makes this investigation safe.

.....
 [1]

(b) When the piece of paper stops burning, the temperature in the jar decreases.

The air pressure in the jar decreases.

Use the particle theory to explain why.

.....

 [2]

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The Periodic Table of Elements

Group									
1	2						7	8	
								2	He helium 4



Science

Stage 8

Paper 1

2023

Cambridge Lower Secondary Progression Test

Mark Scheme

General guidelines on marking

Many descriptive answers can be expressed in a variety of ways. Professional judgement can be used in these cases, providing it matches the marking points and further information in the mark scheme.

Answers may have words spelt incorrectly. Credit is normally given for phonetically correct answers, unless the word has a scientifically different meaning. For example, where the answer should be antennae, credit will be given for antenna but not for anthen (too close to anther).

Only the science is being assessed, so answers do not need to be grammatically correct.

Significant figures will be indicated in the question or in the mark scheme.

Unless specified all marking points are independent.

Annotations and abbreviations

/ or	alternate responses for the same marking point
() brackets	the words or units in brackets do not need to be stated, for example, (recycles or releases or provides) minerals = minerals scores the mark
<u>underline</u>	exact word is required
Accept	an acceptable response
Do not accept	indicates an incorrect response that would contradict another otherwise correct alternative
Ignore	indicates an irrelevant answer that is not creditworthy. Full marks can still be achieved even with answers that are ignored.
Note	provides extra information when necessary
ecf	error carried forward; marks are awarded if an incorrect response has been carried forward from earlier working provided the subsequent working is correct
ora	or reverse argument; for example, as mass increases, volume increases could be written as mass decreases, volume decreases

Question	Answer	Marks	Further Information
1	blood component function	2	all three correct = 2 marks one or two correct = 1 mark if two lines from one blood component and one is incorrect = 0 marks for that blood component

Question	Answer	Marks	Further Information
2(a)	nucleus	1	Ignore protons or neutrons Accept protons and neutrons
2(b)	protons and neutrons	1	both needed any order for the mark
2(c)	Rutherford	1	Accept Bohr
2(d)	negative or minus or –ve	1	Accept 1– or –1 or minus 1
2(e)	(idea that) positive or charge or protons or nucleus and negative charge on electrons (electrostatic) attraction	2	each correct answer = 1 mark Accept electrostatic attraction between protons and electrons = 2 marks

Question	Answer	Marks	Further Information
3	(The mistake at B) the angle of incidence does not equal the angle of reflection or the angles each side of the normal are not equal or it does not obey the law of reflection. (The mistake at D) there is refraction or the ray of light escapes or the ray of light leaves the optical fibre. (The mistake at F) the (reflected) ray of light is not straight or the (reflected) ray of light bends or the (reflected) ray of light curves.	3	each correct answer = 1 mark Accept dotted line for normal

Question	Answer	Marks	Further Information
4(a)		1	Accept any indication of the correct answer, e.g. tick or cross, but circle takes precedence Accept circle drawn in any one of the three areas shown Accept circle drawn in the correct place on the x-axis if more than one circle drawn and any are incorrect = 0 marks
4(b)	cyclic or fluctuating (on a regular basis)	1	Accept constantly changing Accept goes up and down
4(c)	carbon dioxide or methane	1	Accept CO₂ or CH₄ Do not accept CO or CO² Accept other greenhouse gases such as sulfur (VI) fluoride or oxides of nitrogen or water vapour

Question	Answer	Marks	Further Information
5(a)(i)	diffusion	1	Accept gas exchange Ignore absorption
5(a)(ii)	carbon dioxide passes from the blood into the alveolus (air sac)	1	Accept lung(s) for alveolus (air sac)
5(b)	any number greater than 0.04 for carbon dioxide and any number lower than 21.00 for oxygen total for all gases breathed out adds to 100%	2	each correct answer = 1 mark

Question	Answer	Marks	Further Information
6(a)	magnetic field (lines) direction	2	each correct answer = 1 mark Accept direction of magnetic field = 2 marks
6(b)	N S	1	more than one answer circled = 0 marks Accept any indication of the correct answer, e.g. ticking or underlining, but circling takes precedence

Question	Answer	Marks	Further Information
7(a)	(paper) chromatography	1	
7(b)	brown and purple (idea that) these colours have more than one 'dot' in the results	2	each correct answer = 1 mark both needed any order for the mark
7(c)	red and blue	1	both needed any order for the mark Accept any indication of the correct answer, e.g. circling or underlining, but answer line takes precedence

Question	Answer	Marks	Further Information
8	the needle is magnetised/magnetic or the (steel) needle is magnetic (idea that) so it lines up with the Earth's magnetic field	2	each correct answer = 1 mark

Question	Answer	Marks	Further Information
9(a)	(pure water) box contains only $\bigcirc$ molecules (impure water) box contains $\bigcirc$ and at least one other type of molecule	2	each correct diagram = 1 mark Accept particle arrangement for solid, liquid or gas Accept particle arrangement for solid, liquid or gas Note for impure water, some of the molecules must not be linked for the mark
9(b)	analogy	1	more than one answer circled = 0 marks Accept any indication of the correct answer, e.g. ticking or underlining, but circling takes precedence

Question	Answer	Marks	Further Information
10(a)	banana	1	
10(b)(i)	ball and socket	1	
10(b)(ii)	(idea that) one muscle straightens the arm and the other bends the arm	1	Accept one muscle contracts and the other muscle relaxes Accept muscles can only contract or muscles can only pull a bone
10(c)	Lily (no mark) and more oxygen more respiration (to provide energy)	2	Note if incorrect name given = 0 marks Note if no name given = max 1 mark each correct explanation = 1 mark Ignore breathing

Question	Answer	Marks	Further Information																
11(a)	force multiplied by distance	1	Accept f x d or f x s																
11(b)(i)	<table border="1"> <thead> <tr> <th>(distance A in cm)</th><th>(force C in N)</th><th>(distance B in cm)</th><th>force D in N</th></tr> </thead> <tbody> <tr> <td>6</td><td>4</td><td>12</td><td>2</td></tr> <tr> <td>5</td><td>4</td><td>2</td><td>10</td></tr> <tr> <td>4</td><td>2</td><td>8</td><td>1</td></tr> </tbody> </table>	(distance A in cm)	(force C in N)	(distance B in cm)	force D in N	6	4	12	2	5	4	2	10	4	2	8	1	3	correct heading including unit in heading = 1 mark Do not accept units in the body of table all three rows correct = 2 marks one or two rows correct = 1 mark Note rows can be in any row position in the table
(distance A in cm)	(force C in N)	(distance B in cm)	force D in N																
6	4	12	2																
5	4	2	10																
4	2	8	1																
11(b)(ii)	yes (no mark) and (idea that) the force x distance on each side of the ruler are always equal or no (no mark) and (idea that each row of) measurements have not been repeated	1																	

Question	Answer	Marks	Further Information				
12(a)	<table><tr><td>renewable</td><td>non-renewable</td></tr><tr><td>bioplastics tidal power vegetable oil solar power</td><td>fossil fuels</td></tr></table>	renewable	non-renewable	bioplastics tidal power vegetable oil solar power	fossil fuels	2	all five correct = 2 marks two, three or four correct = 1 mark one correct = 0 marks Accept answers in any order in the correct column
renewable	non-renewable						
bioplastics tidal power vegetable oil solar power	fossil fuels						
12(b)	(wind power) is renewable (wind power) will not run out	2	each correct answer = 1 mark Note assume answer refers to wind power unless otherwise specified Accept ora if coal is specified As extra marking points: Accept it is a clean resource or it is a green resource = 1 mark Accept wind is 'free' = 1 mark Accept no pollutants are produced during electricity generation = 1 mark Accept all countries have wind (power) available and some countries do not have coal.				

Question	Answer	Marks	Further Information
13(a)	(dragonflies) decrease in number because the quolls have less food to eat so the quolls eat more dragonflies or (dragonflies) increase in number because there are more insects because there are fewer lizards and small birds	2	each correct answer for decrease = 1 mark Accept (dragonflies) decrease in numbers because the quolls have fewer food sources to eat or Accept quolls eat more insects so there is less food for the dragonflies each correct answer for increase = 1 mark
13(b)	bioaccumulation of toxin in quolls or lizards, birds and dragonflies eat poisoned insects and poison concentrates in the quoll	1	Accept quoll eats many insects and poison accumulates in quoll

Question	Answer	Marks	Further Information
14(a)	any one from does not touch the flame or fire or burning paper keeps fingers away from the balloon being pulled into the jar does not spill water on the floor when filling the balloon does not let the students too close or uses a screen	1	Accept any appropriate safety precaution Ignore wear an apron
14(b)	particles are moving slower or particles have less energy fewer collisions (with surface)	2	each correct answer = 1 mark

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Science

Stage 8

Paper 2

2023

Cambridge Lower Secondary Progression Test

Name

Class

Date

45 minutes

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Write your answer to each question in the space provided.
- You should show all your working on the question paper.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

- 1 The table shows information about four different adults who play sports.

The adults are the same age and the same height.

adults	physical activity level	do they smoke?
W	low	no
X	high	no
Y	high	yes
Z	low	yes

- (a) (i) Which adult, **W**, **X**, **Y** or **Z**, is most likely to have health problems?

.....

[1]

- (ii) The four adults test their fitness by exercising.

After exercising, they measure their pulse rate.

What do they do to make the test fair?

Tick (✓) the correct answer.

two adults run and two adults walk for 5 minutes ☐

use a ruler to measure distance ☐

all four adults run at the same speed for 1 minute ☐

repeat the exercise four times ☐

[1]

- (b) (i) Adult **Y** takes a vitamin A tablet every morning.

What is the function of vitamin A?

..... [1]

- (ii) Adult **Z** eats a diet that has a very high fat content.

Write down **one** health problem caused by this type of diet.

..... [1]

- 2 Aiko reacts iron with dilute sulfuric acid.

She takes the temperature of the dilute sulfuric acid before she adds the iron.

She takes the temperature of the reaction mixture at the end of the reaction.

Look at her results.

temperature of dilute sulfuric acid at start = 18°C

temperature of reaction mixture at end = 28°C

- (a) Write down the **type** of reaction between iron and dilute sulfuric acid.

Choose from the list.

endothermic

exothermic

neutralisation

precipitation

type of reaction

Explain your answer.

.....

.....

[2]

- (b) Iron reacts with sulfuric acid.

Iron sulfate and hydrogen are made.

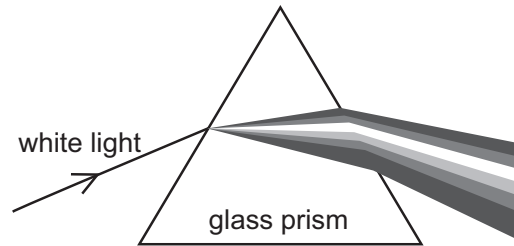
Write down the **word** equation for this reaction.

.....

[1]

3 White light is made of many colours.

(a) Oliver uses a glass prism and a ray of white light.



Complete the sentences.

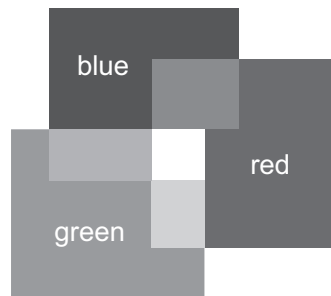
A ray of white light enters the glass prism.

This ray of white light changes speed because the light is

White light is split into different colours by the process of

[2]

(b) Oliver adds different colours of light together.



Complete the additions of light.

One has been done for you.

red + blue = magenta

red + green =

green + blue =

red + blue + green =

[2]

4 Name **two** objects that are found in all known galaxies.

1

2

[2]

5 The picture shows a woodland ecosystem.



(a) One habitat shown in the woodland ecosystem is tree roots.

Name **two other** habitats shown in the picture.

1

2

[2]

(b) Jamila observes a new species of bird in the woodland.

She makes a hypothesis,

‘The new species of bird reduces the number of other species of bird in the woodland.’

Jamila counts the number of species of bird in the woodland.

Six months later, she counts the number of species of bird in the same place in the woodland.

- (i) Write down **one other** factor she must keep the same during both counts.

..... [1]

- (ii) What is the dependent variable?

..... [1]

- (iii) Look at Jamila's table of results.

count number	number of species of bird
1	14
2	12

Do these results support her hypothesis?

.....

Give **two** reasons for your answer.

1

.....

2

.....

[2]

- 6 Look at the table showing the reactions of some metals with cold water and with dilute hydrochloric acid.

metal	reaction with cold water	reaction with dilute hydrochloric acid
A	no reaction	no reaction
B	floats on water, reacts quickly and gives off a gas	floats on acid, reacts very quickly and gives off a gas
C	reacts slowly and gives off a gas	reacts quickly and gives off a gas
D	no reaction	reacts slowly and gives off a gas

Look at this list of metals.

calcium

gold

magnesium

sodium

zinc

- (a) Identify the metals **A**, **B**, **C** and **D**.

Choose from the list.

Metal **A** is

Metal **B** is

Metal **C** is

Metal **D** is

[3]

- (b) Write down the name of the gas made when metals **B** and **C** react with water.

.....

[1]

- 7 Lily uses this equipment to make an electromagnet.



battery



copper wire



iron nail

- (a) Describe how Lily makes an electromagnet.

.....

.....

..... [2]

- (b) Which of these changes **increase** the strength of her electromagnet?

Tick (✓) the **two** correct answers.

cover the copper wire in plastic

☐

use a non metal for the wire

☐

use two batteries

☐

remove the iron nail

☐

increase the current

☐

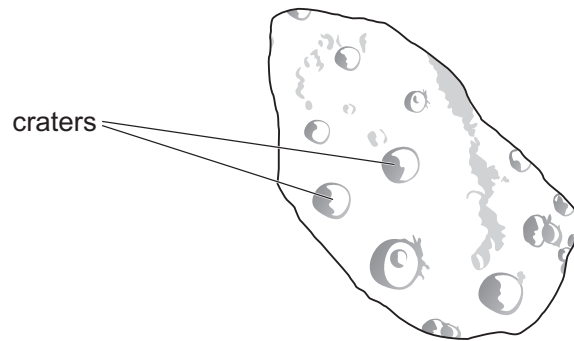
connect the equipment for a shorter time

☐

[2]

- 8 The picture shows an asteroid.

Asteroids are found in space.



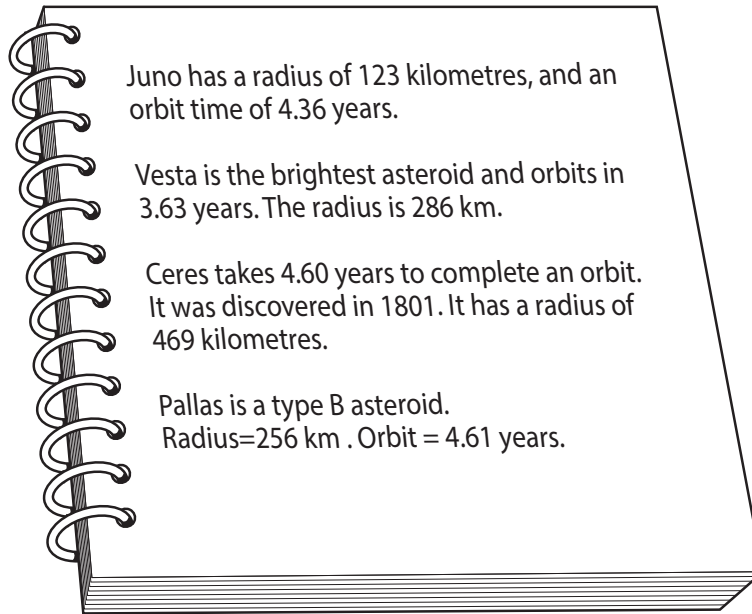
- (a) (i) What is an asteroid?

.....
..... [1]

- (ii) Suggest how craters are made on asteroids.

.....
..... [1]

(b) Pierre writes down some information about four asteroids.



He draws a table to show the information more clearly.

(i) Present his information on orbit time and radius for each asteroid in the table.

[3]

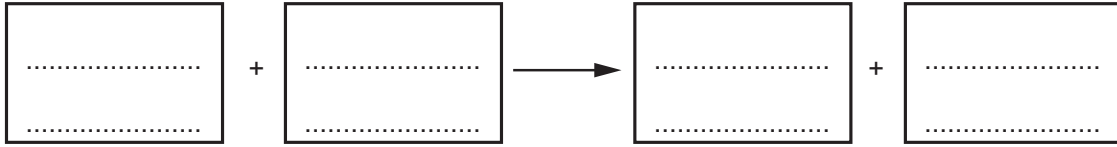
(ii) All four asteroids travel the same distance in one year.

Which asteroid travels at the **greatest** mean speed?

..... [1]

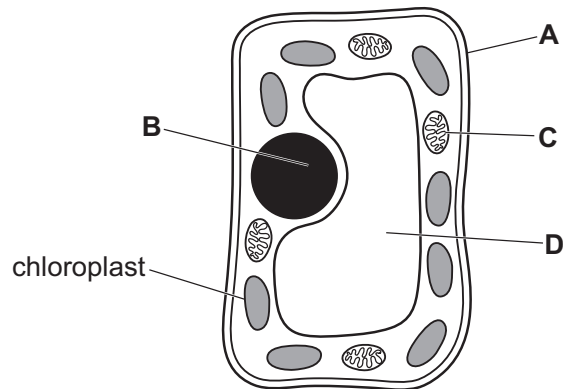
9 This question is about aerobic respiration.

(a) Write down the **word** equation for aerobic respiration.



[2]

(b) Carlos draws a diagram of a plant cell.



Where does aerobic respiration happen in this cell?

Choose from the letters **A**, **B**, **C** or **D**.

letter

[1]

10 Blessy dissolves salt in water to make a salt solution.

She makes four solutions.

Look at the quantities she uses.

solution	mass of salt in g	volume of water in cm ³
A	1	10
B	6	30
C	5	20
D	10	50

(a) Which solution is the **most** concentrated?

Circle the correct answer.

A

B

C

D

Explain your answer.

.....

[2]

(b) Describe how the solubility of salt changes as the temperature increases.

..... [1]

11 Ahmed has time and distance measurements for a bicycle journey.

time in s	distance in m
0	0
20	60
40	240
60	420
80	600
100	760

(a) Write down the equipment he uses to make these measurements.

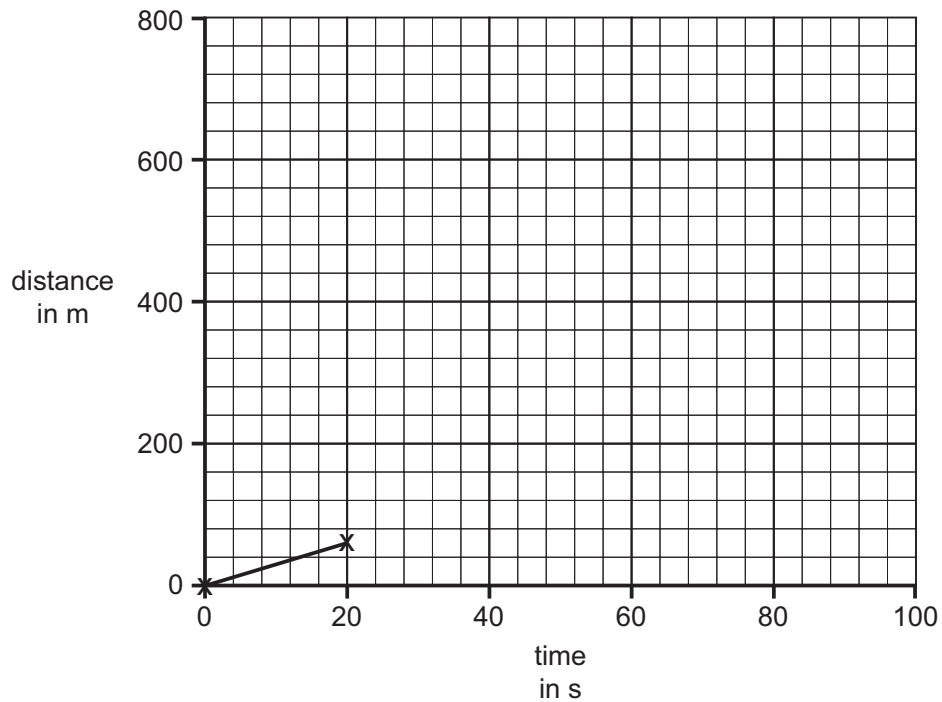
time to measure 20 s

distance to measure 60 m

[2]

(b) Complete the graph by:

- plotting the other four points
- drawing a line to connect the points.



[2]

(c) Calculate the speed when a distance of 60 m is travelled in 20 s.

Use the equation shown.

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

speed = m/s
[1]

12 Here are some sentences about climate and weather.

Tick (✓) each correct sentence.

Climate is the day to day changes in the atmosphere.

☐

Climate changes over a long period of time.

☐

Weather changes over a short period of time.

☐

Weather gives information about the atmosphere.

☐

[2]

13 Priya uses chemical formulae to model chemical compounds.

The chemical formula of sodium carbonate is Na_2CO_3 .

The formula of sodium carbonate contains:

- 2 atoms of sodium, Na
- 1 atom of carbon, C
- 3 atoms of oxygen, O.

(a) The formula of carbon dioxide contains:

- 1 atom of carbon, C
- 2 atoms of oxygen, O.

Write down the chemical formula for carbon dioxide.

formula

[1]

(b) The formula of nitric acid contains:

- 1 atom of hydrogen, H
- 1 atom of nitrogen, N
- 3 atoms of oxygen, O.

Write down the chemical formula of nitric acid.

formula

[1]

(c) Look at the word equation.



The formula of sodium nitrate is NaNO_3 .

The formula of water is H_2O .

Write down the **symbol** equation for this reaction.

..... [2]

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Group																		
1	2	Key						3	4	5	6	7	8					
		1 H hydrogen 1														2 He helium 4		
3 Li lithium 7	4 Be beryllium 9	atomic number atomic symbol name relative atomic mass										5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	Ne neon 20	
11 Na sodium 23	12 Mg magnesium 24											13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40	
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84	
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131	
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids		72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —
87 Fr francium —	88 Ra radium —	89–103 actinoids		104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	113 Nh nihonium —	114 Fl flerovium —	115 Mc moscovium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganesson —

lanthanoids	57	La	lanthanum	139	58	Ce	cerium	140	59	Pr	praseodymium	141	60	Nd	neodymium	144	61	Pm	promethium	—	62	Sm	samarium	150	63	Eu	europium	152	64	Gd
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Science

Stage 8

Paper 2

2023

Cambridge Lower Secondary Progression Test

Mark Scheme

General guidelines on marking

Many descriptive answers can be expressed in a variety of ways. Professional judgement can be used in these cases, providing it matches the marking points and further information in the mark scheme.

Answers may have words spelt incorrectly. Credit is normally given for phonetically correct answers, unless the word has a scientifically different meaning. For example, where the answer should be antennae, credit will be given for antenna but not for anthen (too close to anther).

Only the science is being assessed, so answers do not need to be grammatically correct.

Significant figures will be indicated in the question or in the mark scheme.

Unless specified all marking points are independent.

Annotations and abbreviations

/ or	alternate responses for the same marking point
() brackets	the words or units in brackets do not need to be stated, for example, (recycles or releases or provides) minerals = minerals scores the mark
<u>underline</u>	exact word is required
Accept	an acceptable response
Do not accept	indicates an incorrect response that would contradict another otherwise correct alternative
Ignore	indicates an irrelevant answer that is not creditworthy. Full marks can still be achieved even with answers that are ignored.
Note	provides extra information when necessary
ecf	error carried forward; marks are awarded if an incorrect response has been carried forward from earlier working provided the subsequent working is correct
ora	or reverse argument; for example, as mass increases, volume increases could be written as mass decreases, volume decreases

Question	Answer	Marks	Further Information
1(a)(i)	Z	1	
1(a)(ii)	run at the same speed for 1 minute	1	more than one answer ticked = 0 marks Accept any indication of the correct answer, e.g. circling or underlining, but ticking takes precedence
1(b)(i)	(improves) eyesight or vision (in low light conditions)	1	Accept for healthy skin or (improves) immune system or cell growth
1(b)(ii)	obesity or heart disease or increase in (bad) cholesterol or high blood pressure or stroke	1	

Question	Answer	Marks	Further Information
2(a)	exothermic (idea of) temperature increase	2	each correct answer = 1 mark Accept any indication of the correct answer, e.g. circling or underlining, but answer line takes precedence
2(b)	iron + sulfuric acid → iron sulfate + hydrogen	1	reactants correct in either order and products correct in either order = 1 mark Ignore dilute Accept correct formulae, but name takes precedence $\text{Fe} + \text{H}_2\text{SO}_4 \rightarrow \text{FeSO}_4 + \text{H}_2$ Accept = instead of →

Question	Answer	Marks	Further Information
3(a)	refracted dispersion	2	each correct answer = 1 mark Ignore refraction in second sentence
3(b)	(red + green =) yellow (green + blue =) cyan (red + blue + green =) white	2	all three correct = 2 marks one or two correct = 1 mark

Question	Answer	Marks	Further Information
4	any two from (stellar) gas (stellar) dust stars planets moons asteroids comets nebulae	2	each correct answer = 1 mark Accept any other suitable objects found in all known galaxies, e.g. black hole or meteorites Accept planetary systems if planets and moons have not been given

Question	Answer	Marks	Further Information
5(a)	any two from fallen tree stream or river or water fallen leaves tree canopy or treetops or branches of the tree	2	each correct answer = 1 mark Accept any suitable habitat shown in the picture, e.g. grass
5(b)(i)	length of time she looks for or time of day	1	Accept any suitable factor that must be kept the same, e.g. temperature or weather
5b(ii)	number of species of bird	1	
5b(iii)	yes or no or maybe (no mark) any two explanations from the results are very close to each other number of species (of bird) has decreased may have misidentified species must identify if the new species is still present should sample over a longer time interval may have looked for species for different lengths of time there may be migration into or out of the woodland there may be seasonal differences not sampled enough places in the woodland	2	Note the explanations must match the decision about the hypothesis each correct explanation = 1 mark if no decision about the hypothesis but two correct explanations = 1 mark

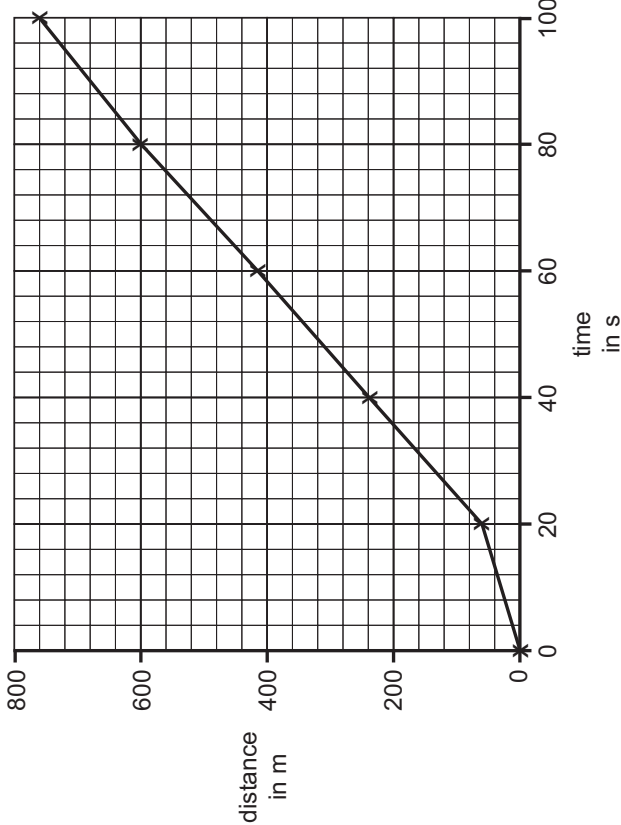
Question	Answer	Marks	Further Information
6(a)	(Metal A is) gold (Metal B is) sodium (Metal C is) calcium (Metal D is) zinc	3	all four correct = 3 marks three correct = 2 marks one or two correct = 1 mark Accept metal C is magnesium
6(b)	hydrogen	1	Accept H ₂

Question	Answer	Marks	Further Information
7(a)	(idea of) a complete circuit (idea of) coil the wire around the nail	2	each correct answer = 1 mark Accept answer in the form of a diagram
7(b)	cover the copper wire in plastic use a non metal for the wire use two batteries remove the iron nail increase the current connect the equipment for a shorter time	2	each correct tick = 1 mark one incorrect tick = maximum 1 mark two incorrect ticks = 0 marks Accept any indication of the correct answer, e.g. circling or underlining, but ticking takes precedence

Question	Answer	Marks	Further Information															
8(a)(i)	(small) rock in space	1	Accept minor planet or planetoid for rock Accept (small) rock orbiting the Sun															
8(a)(ii)	collisions with other asteroids or dust or objects	1																
8(b)(i)	<table><thead><tr><th>asteroid or name</th><th>(orbit) time in years</th><th>radius in kilometres</th></tr></thead><tbody><tr><td>Vesta</td><td>3.63</td><td>286</td></tr><tr><td>Ceres</td><td>4.60</td><td>469</td></tr><tr><td>Juno</td><td>4.36</td><td>123</td></tr><tr><td>Pallas</td><td>4.61</td><td>256</td></tr></tbody></table>	asteroid or name	(orbit) time in years	radius in kilometres	Vesta	3.63	286	Ceres	4.60	469	Juno	4.36	123	Pallas	4.61	256	3	correct headings including unit in headings = 1 mark all four rows (in any order) correct = 2 marks one, two or three rows (in any order) correct = 1 mark Do not accept units in the body of table Accept km for kilometres Note columns can be in any order Ignore additional columns
asteroid or name	(orbit) time in years	radius in kilometres																
Vesta	3.63	286																
Ceres	4.60	469																
Juno	4.36	123																
Pallas	4.61	256																
8(b)(ii)	Vesta	1	Accept ecf from 8(b)(i) for whichever asteroid is shown as completing its orbit in the shortest time															

Question	Answer	Marks	Further Information
9(a)	glucose + oxygen → carbon dioxide + water	2	reactants correct in either order = 1 mark products correct in either order = 1 mark Accept correct formulae, but name takes precedence $C_6H_{12}O_6 + O_2 \rightarrow CO_2 + H_2O$ Accept sugar for glucose
9(b)	C	1	more than one letter on answer line = 0 marks Accept any indication of the correct answer, e.g. circling, but answer line takes precedence

Question	Answer	Marks	Further Information
10(a)	C (idea of) greatest mass of salt per cm³ of water	2	each correct answer = 1 mark more than one answer circled = 0 marks if A, B or D indicated = 0 marks for the question Accept any indication of the correct answer, e.g. ticking or underlining, but circling takes precedence Accept any calculation that demonstrates greatest mass of salt per cm³ of water e.g. $1/10 = 0.1$, $6/30 = 0.2$, $5/20 = 0.25$, $10/50 = 0.2$
10(b)	(idea that the solubility) increases	1	

Question	Answer	Marks	Further Information
11(a)	(time to measure 20 s) stop-watch or timer or stop-clock (distance to measure 60 m) measuring tape	2	both answers correct for the mark Ignore watch or clock or chronometer Accept trundle wheel Ignore meter rule(r)
11(b)		2	all four points plotted correctly = 1 mark dot-to-dot line drawn = 1 mark Accept ecf for dot-to-dot line from incorrectly plotted points Accept plots $\pm 1\frac{1}{2}$ small square
11(c)	3 (m / s)	1	

Question	Answer	Marks	Further Information
12	Weather changes over a short period of time. Climate changes over a long period of time.	2	two correct ticks = 2 marks one correct tick = 1 mark three ticks and two correct = 1 mark three ticks and one correct = 0 marks Accept any indication of the correct answer, e.g. circling or underlining, but ticking takes precedence

Question	Answer	Marks	Further Information
13(a)	CO ₂	1	Accept O ₂ C
13(b)	HNO ₃	1	Accept O ₃ NH or O ₃ HN or NO ₃ H or HO ₃ N or NHO ₃
13(c)	Na ₂ CO ₃ + HNO ₃ → NaNO ₃ + CO ₂ + H ₂ O	2	reactants correct in either order = 1 mark products correct in any order = 1 mark Accept ecf from parts (a) and (b) Accept = instead of → Ignore any attempt to balance the equation

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